Band-1 Hadamard order 14084 – Classical Paley-I (q=14083); IN SPK via Paley-I; Gap 2052 Still OPEN

H_14084 Paley-I PASS; Paley-II FAIL/SKIP (q=7041=3*2347 NOT prime; ===1 mod 4 but composite); Kronecker FAIL/SKIP (4 3521; 3521===1 mod 4 NOT Hadamard-admissible; 3521=7*503 (odd composite)); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-10-08

Document class. Scientific computational / construction certificate (Band-1).

Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.

Parents. `2026-10-08_Band1_Hadamard_14072_PaleyI`; `2026-10-08_Band1_Hadamard_14052_PaleyI`; `2026-10-08_Band1_Hadamard_14012_PaleyI`; `2026-10-07_Band1_Hadamard_14000_PaleyI`; `2026-10-07_Band1_Hadamard_13968_PaleyI`; `2026-10-07_Band1_Hadamard_13964_PaleyI`; `2026-10-07_Band1_Hadamard_13932_PaleyI`; `2026-10-07_Band1_Hadamard_13908_PaleyI`; `2026-10-07_Band1_Hadamard_13904_PaleyI`; `2026-10-07_Band1_Hadamard_13884_PaleyI`; `2026-10-07_Band1_Hadamard_13880_PaleyI`; `2026-10-07_Band1_Hadamard_13860_PaleyI`; `2026-10-07_Band1_Hadamard_13832_PaleyI`; `2026-10-07_Band1_Hadamard_13808_PaleyI`; `2026-10-07_Band1_Hadamard_13800_PaleyI`; `2026-10-07_Band1_Hadamard_13764_PaleyI`; `2026-10-07_Band1_Hadamard_13760_PaleyI`; `2026-10-07_Band1_Hadamard_13752_PaleyI`; `2026-10-07_Band1_Hadamard_13724_PaleyI`; `2026-10-07_Band1_Hadamard_13712_PaleyI`; `2026-10-07_Band1_Hadamard_13692_PaleyI`; `2026-10-07_Band1_Hadamard_13688_PaleyI`; `2026-10-06_Band1_Hadamard_13680_PaleyI`; `2026-10-06_Band1_Hadamard_13628_PaleyI`; `2026-10-06_Band1_Hadamard_13620_PaleyI`; `2026-10-06_Band1_Hadamard_13592_PaleyI`; `2026-10-06_Band1_Hadamard_13568_PaleyI`; `2026-10-06_Band1_Hadamard_13524_PaleyI`; `2026-10-06_Band1_Hadamard_13500_PaleyI`; `2026-10-06_Band1_Hadamard_13488_PaleyI`; `2026-10-06_Band1_Hadamard_13464_PaleyI`; `2026-10-06_Band1_Hadamard_13452_PaleyI`; `2026-10-06_Band1_Hadamard_13412_PaleyI`; `2026-10-06_Band1_Hadamard_13400_PaleyI`; `2026-10-06_Band1_Hadamard_13368_PaleyI`; `2026-10-06_Band1_Hadamard_13340_PaleyI`; `2026-10-06_Band1_Hadamard_13332_PaleyI`; `2026-10-06_Band1_Hadamard_13328_PaleyI`; `2026-10-06_Band1_Hadamard_13292_PaleyI`; `2026-10-06_Band1_Hadamard_13268_PaleyI`; `2026-10-06_Band1_Hadamard_13260_PaleyI`; `2026-10-06_Band1_Hadamard_13220_PaleyI`; `2026-10-06_Band1_Hadamard_13188_PaleyI`; `2026-10-06_Band1_Hadamard_13184_PaleyI`; `2026-10-06_Band1_Hadamard_13172_PaleyI`; `2026-10-05_Band1_Hadamard_13164_PaleyI`; `2026-10-05_Band1_Hadamard_13160_PaleyI`; `2026-10-05_Band1_Hadamard_13152_PaleyI`;

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2026-10-05_Band1_Hadamard_13008_PaleyI; 2026-10-05_Band1_Hadamard_13004_PaleyI;
2026-10-05_Band1_Hadamard_12984_PaleyI; 2026-10-05_Band1_Hadamard_12980_PaleyI;
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2026-10-05_Band1_Hadamard_12912_PaleyI; 2026-10-05_Band1_Hadamard_12908_PaleyI;
2026-10-04_Band1_Hadamard_12900_PaleyI; 2026-10-04_Band1_Hadamard_12824_PaleyI;
2026-10-04_Band1_Hadamard_12800_PaleyI; 2026-10-04_Band1_Hadamard_12792_PaleyI;
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2026-10-03_Band1_Hadamard_12264_PaleyI; 2026-10-03_Band1_Hadamard_12252_PaleyI;
2026-10-03_Band1_Hadamard_12240_PaleyI; 2026-10-03_Band1_Hadamard_12228_PaleyI;
2026-10-03_Band1_Hadamard_12212_PaleyI; 2026-10-03_Band1_Hadamard_12204_PaleyI;
2026-10-03_Band1_Hadamard_12164_PaleyI; 2026-10-03_Band1_Hadamard_12144_PaleyI;
2026-10-03_Band1_Hadamard_12120_PaleyI; 2026-10-03_Band1_Hadamard_12108_PaleyI;
2026-10-03_Band1_Hadamard_12072_PaleyI; 2026-10-03_Band1_Hadamard_12044_PaleyI;
2026-10-03_Band1_Hadamard_12012_PaleyI; 2026-10-03_Band1_Hadamard_12008_PaleyI;
2026-10-03_Band1_Hadamard_11988_PaleyI; 2026-10-03_Band1_Hadamard_11972_PaleyI;
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2026-10-03_Band1_Hadamard_11928_PaleyI; 2026-10-02_Band1_Hadamard_11924_PaleyI;
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2026-09-30_Band1_Hadamard_11448_PaleyI; 2026-09-30_Band1_Hadamard_11444_PaleyI;

2026-09-30_Band1_Hadamard_11424_PaleyI; 2026-09-30_Band1_Hadamard_11412_PaleyI;
2026-09-30_Band1_Hadamard_11400_PaleyI; 2026-09-30_Band1_Hadamard_11384_PaleyI;
2026-09-30_Band1_Hadamard_11352_PaleyI; 2026-09-30_Band1_Hadamard_11312_PaleyI;
2026-09-30_Band1_Hadamard_11300_PaleyI; 2026-09-30_Band1_Hadamard_11288_PaleyI;
2026-09-30_Band1_Hadamard_11280_PaleyI; 2026-09-30_Band1_Hadamard_11252_PaleyI;
2026-09-29_Band1_Hadamard_11244_PaleyI; 2026-09-29_Band1_Hadamard_11240_PaleyI;
2026-09-29_Band1_Hadamard_11172_PaleyI; 2026-09-29_Band1_Hadamard_11160_PaleyI;
2026-09-29_Band1_Hadamard_11132_PaleyI; 2026-09-29_Band1_Hadamard_11120_PaleyI;
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2026-09-29_Band1_Hadamard_11004_PaleyI; 2026-09-29_Band1_Hadamard_10988_PaleyI;
2026-09-29_Band1_Hadamard_10980_PaleyI; 2026-09-29_Band1_Hadamard_10940_PaleyI;
2026-09-29_Band1_Hadamard_10892_PaleyI; 2026-09-29_Band1_Hadamard_10884_PaleyI;
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2026-09-29_Band1_Hadamard_10772_PaleyI; 2026-09-29_Band1_Hadamard_10740_PaleyI;
2026-09-29_Band1_Hadamard_10724_PaleyI; 2026-09-29_Band1_Hadamard_10712_PaleyI;
2026-09-28_Band1_Hadamard_10692_PaleyI; 2026-09-28_Band1_Hadamard_10688_PaleyI;
2026-09-28_Band1_Hadamard_10668_PaleyI; 2026-09-28_Band1_Hadamard_10652_PaleyI;
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2026-09-28_Band1_Hadamard_10628_PaleyI; 2026-09-28_Band1_Hadamard_10608_PaleyI;
2026-09-28_Band1_Hadamard_10568_PaleyI; 2026-09-28_Band1_Hadamard_10560_PaleyI;
2026-09-28_Band1_Hadamard_10532_PaleyI; 2026-09-28_Band1_Hadamard_10500_PaleyI;
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2026-09-28_Band1_Hadamard_10400_PaleyI; 2026-09-24_Band1_Hadamard_10392_PaleyI;
2026-09-24_Band1_Hadamard_10344_PaleyI; 2026-09-24_Band1_Hadamard_10332_PaleyI;
2026-09-24_Band1_Hadamard_10304_PaleyI; 2026-09-24_Band1_Hadamard_10272_PaleyI;
2026-09-23_Band1_Hadamard_10268_PaleyI; 2026-09-23_Band1_Hadamard_10224_PaleyI;
2026-09-23_Band1_Hadamard_10212_PaleyI; 2026-09-23_Band1_Hadamard_10164_PaleyI;
2026-09-23_Band1_Hadamard_10160_PaleyI; 2026-09-23_Band1_Hadamard_10152_PaleyI;
2026-09-23_Band1_Hadamard_10140_PaleyI; 2026-09-23_Band1_Hadamard_10112_PaleyI;
2026-09-23_Band1_Hadamard_10104_PaleyI; 2026-09-23_Band1_Hadamard_10100_PaleyI;
2026-09-23_Band1_Hadamard_10068_PaleyI.

Lab. `/workspace/experiments/band1/hadamard_14084_paleyI/`.

Cutoff. 2026-10-08T11:28:04Z (~7:28AM ET).

Honesty stamp (non-negotiable).

This memo claims **finite construction of order 14084 only (verified max|HHˆT-nI|=0), via classical Paley-I q=14083 (prime ===3 mod 4). 14084 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 14084:

Paley-I **PASS** ($q = 14083$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** (q=7041=3*2347 NOT prime; ===1 mod 4 but composite); Kronecker **FAIL/SKIP** (note $14084 = 4 \cdot 3521$; $3521 \equiv 1 \pmod 4$ NOT Hadamard-admissible; $3521 = 7 \cdot 503$ (odd composite)). **14084** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **14072+14052+14012+14000+13968+13964+13932+13908+13904+13884+13880+13860+13832+138** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

ewpage

Abstract

Finite construction of order **14084**.

- H_{14084} = Paley-I ($q = 14083$) – classical SPK
- Paley-II for 14084: **FAIL/SKIP** ($q = 7041 = 3 \cdot 2347$ NOT prime; $\equiv 1 \pmod 4$ but composite)
- Kronecker for 14084: **FAIL/SKIP** ($4 \cdot 3521$; $3521 \equiv 1 \pmod 4$ NOT Hadamard-admissible; $3521 = 7 \cdot 503$ (odd composite))
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

```
python3 verify_hadamard_14084.py
```

Results. overall_pass=True; units 644/644; row blocks 111/111 (block size 128); wall 2690.5577s; constructed max|HHˆT-nI|=0.

Cutoff: 2026-10-08T11:28:04Z (~7:28AM ET)

Arithmetic

- Target $n = 14084 = q + 1$ with $q = 14083$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 7041 = 3 \cdot 2347$ NOT prime; $\equiv 1 \pmod 4$ but composite — FAIL/SKIP.
- Kronecker: $14084 = 4 \cdot 3521$; $3521 \equiv 1 \pmod 4$ NOT Hadamard-admissible; $3521 = 7 \cdot 503$ (odd composite) — FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_14084 Paley-I	Paley-I q=14083	0	2649.6932

Object	Method	max|HHˆT-nI|	Wall (s)
H_14084 Paley-II	FAIL/SKIP (q=7041=3*2347 NOT prime; ===1 mod 4 but composite)	–	–
H_14084 Kronecker	FAIL/SKIP (4*3521; 3521===1 mod 4 NOT Hadamard-admissible; 3521=7503 (odd composite))	–	–

SPK closure note

Seeds include prior SPK through **11528** lineage (**11528**; **11520**; **11504**; **11492**; **11484**; **11472**; **11468**; **11448**; **11444**; **11424**; **11412**; **11400**; **11384**; **11352**; **11312**; **11300**; **11288**; **11280**; **11252**; **11244**; **11240**; **11172**; **11160**; **11132**; **11120**; **11088**; **11084**; **11072**; **11060**; **11048**; **11028**; **11004**; **10988**; **10980**; **10940**; **10904**; **10892**; **10884**; **10860**; **10848**; **10832**; **10800**; **10772**; **10740**; **10724**; **10712**; **10692**; **10688**; **10668**; **10652**; **10640**; **10632**; **10628**; **10608**; **10568**; **10560**; **10532**; **10500**; **10488**; **10464**; **10460**; **10428**; **10400**; **10392**; **10344**; **10332**; **10304**; **10272**; **10268**; **10260**; **10248**; **10244**; **10224**; **10212**; **10164**; **10160**; **10152**; **10140**; **10112**; **10104**; **10100**; **10068**; **10040**; **10008**; **9968**; **9932**; **9924**; **9908**; **9888**; **9884**; **9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; …; **NOT** 8948/8644/8516/…) + verified **10464** (+ prior **10392+10344+10332+10304+10272+10268+10224+10212+10164+10160+10152+10140+10112+101**
Next constructive gap **2052** (OPEN).

Quantity	Value
12488 in SPK	**True** (Paley-I q=14083)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q2=6961->n=13924 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6961 prime ===1 mod 4; Paley-I q=13923=3^2*71317 *NOT prime -> FAIL; Kronecker 13924=4*3481; 3481=59^2 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3481); 13924=2^2*59^2 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6977->n=13956 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6977 prime ===1 mod 4; Paley-I q=13955=5*2791 NOT prime -> FAIL; Kronecker 13956=4*3489; 3489=3*1163 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3489); 13956=2^2*3*1163 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6997->n=13996 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6997 prime ===1 mod 4; Paley-I q=13995=3^2*5*311 NOT prime -> FAIL; Kronecker 13996=4*3499; 3499 odd prime (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3499); 13996=2^2*3499 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7001->n=14004 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7001 prime ===1 mod 4; Paley-I q=14003=11*1967 NOT prime -> FAIL; Kronecker 14004=4*3501; 3501=3^2*389 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3501); 14004=2^2*3^2*389 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7013->n=14028 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7013 prime ===1 mod 4; Paley-I q=14027=13^2*83 *NOT prime -> FAIL; Kronecker 14028=4*3507; 3507=3*7*167 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3507); 14028=2^2*3*7*167 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | FIRST Paley-II DIRECT after 14084: q2=7057->n=14116 (q2=7057 prime ===1 mod 4; Paley-I q=14115=3*5*941 NOT prime -> FAIL; Kronecker 14116=4*3529; 3529 odd prime (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3529); 14116=2^2*3529 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII); Paley-I DIRECT 14088/14108 come first) | Kronecker-only (no direct Paley route; note-only, NOT on box): none in 14084..14088 (no n===0 mod 4 strictly between); beyond: 14096=4*3524 (3524=2^2*881 ===0 mod 4 order-admissible but H_3524 has no direct classical Paley route: 3523=13*271 composite; 1761=3*587 composite), 14112=4*3528 (3528=2^(3^3*27)*2 ===0 mod 4; H_3528 via Paley-I q=3527 prime ===3 mod 4) | next Paley-I DIRECT: q=14087->n=14088 (prime ===3 mod 4; no prime in 14084..14086; FIRST n===0 mod 4 after 14084 with any direct classical Paley route; Paley-II q=7043 prime but ===3 mod 4 (needs ===1) -> FAIL/SKIP; Kronecker 14088=4*3522; 3522=2*3587 ===2 mod 4 -> NOT

*Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3522); 14088=2^3*3*587 ===8 mod 16 = 14072/13880/13832-like n===8 mod 16 / m===2 mod 4 shape (q2===3 mod 4; here 7043 is prime but ===3 mod 4)) (or gap 2052 / Track B)*